Упражнение №8. I част

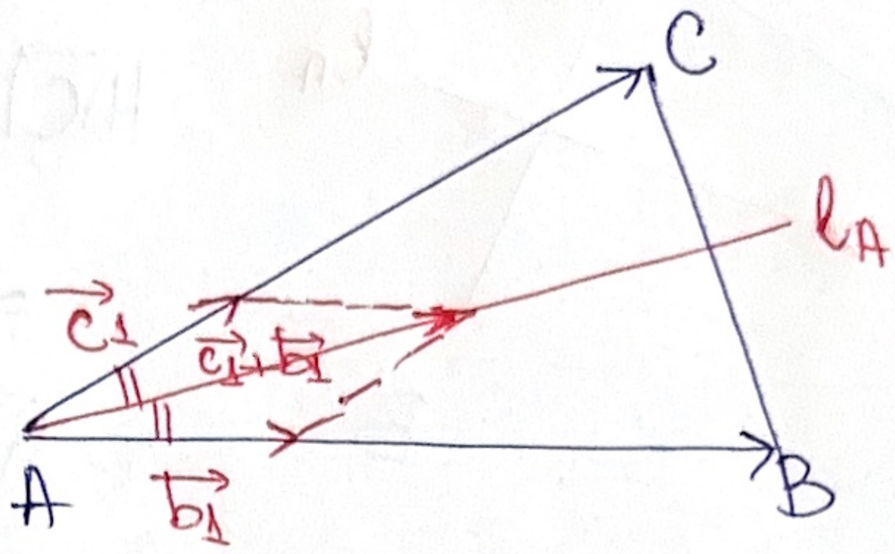
1 зад. OKC Oxy

а) $A(1,2)$ (за упр.)

$B(-1,3)$

$C(5,4)$

Да се намери у-е на вътрешната ъглополовяща l_A при върха A на $\triangle ABC$



реше.: 1) $\overrightarrow{AB} \parallel \overrightarrow{AB}(-2,1) \Rightarrow \vec{b_1} = \dfrac{\overrightarrow{AB}}{|\overrightarrow{AB}|} \Rightarrow \vec{b_1}\left(\dfrac{-2}{\sqrt{5}}, \dfrac{1}{\sqrt{5}}\right)$

$|\overrightarrow{AB}| = \sqrt{(-2)^2+1^2} = \sqrt{5}$

$AC \parallel \overrightarrow{AC}(4,2) \Rightarrow \vec{c_1} = \dfrac{\overrightarrow{AC}}{|\overrightarrow{AC}|} \Rightarrow \vec{c_1}\left(\dfrac{2}{\sqrt{5}}, \dfrac{1}{\sqrt{5}}\right)$

$|\overrightarrow{AC}| = \sqrt{4^2+2^2} = \sqrt{20} = 2\sqrt{5}$

$l_A: \begin{cases} \supset A(1,2) \\ \parallel (\vec{c_1}+\vec{b_1})\left(0, \dfrac{2}{\sqrt{5}}\right) \parallel \vec{e_2}(0,1) \end{cases}$

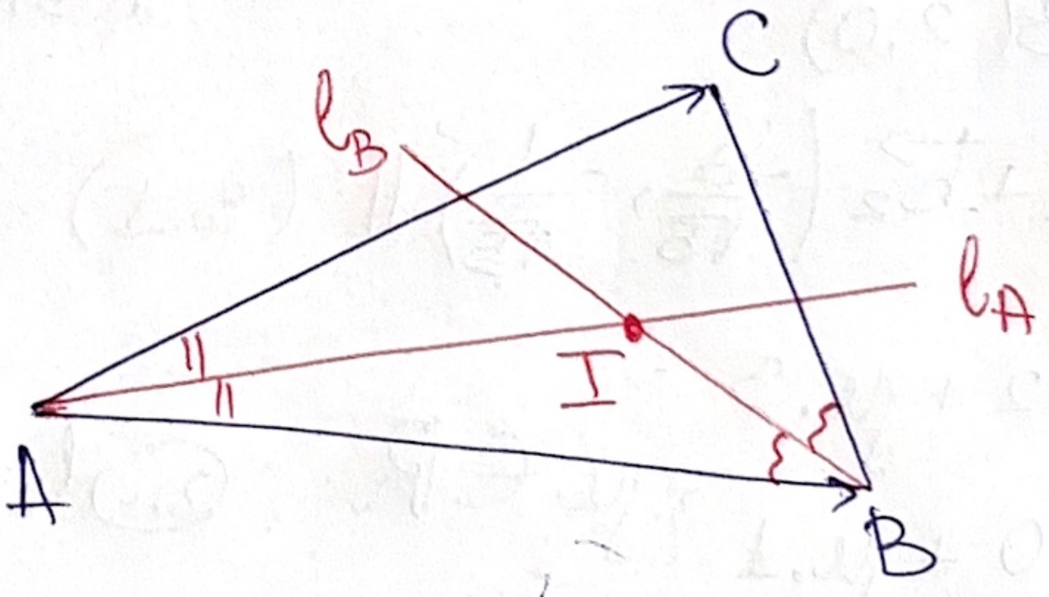
$\Rightarrow l_A: \begin{cases} x = 1+s.0 \\ y = 2+s.1 \end{cases}, s \in \mathbb{R}$ (1н.) (2н.) $l_A: x-1=0$

б) $A(1,-2)$,

$B(2,0)$

$C\left(-\dfrac{2}{3}, \dfrac{4}{3}\right)$

Да се намерят координатите на вписаната в $\triangle ABC$ окръжност и радиуса на тази окр. ($r=?$)

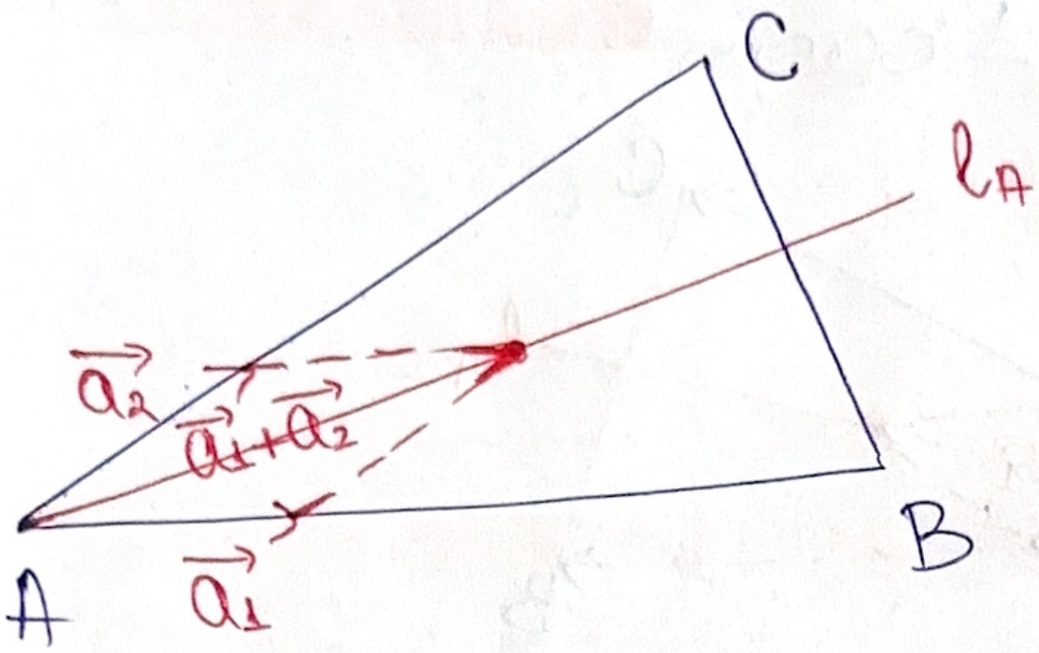


реше.: Нека l_A и l_B са вътрешните ъглополовящи съотв. при върха A и B.

Идея: $l_A \cap l_B = \supset I$

1) $AB \parallel \overrightarrow{AB}(1,2) \Rightarrow \vec{a_1} = \dfrac{\overrightarrow{AB}}{|\overrightarrow{AB}|} \Rightarrow \vec{a_1}\left(\dfrac{1}{\sqrt{5}}, \dfrac{2}{\sqrt{5}}\right)$

$|\overrightarrow{AB}| = \sqrt{1+2^2} = \sqrt{5}$



$AC \parallel \overrightarrow{AC}\left(-\frac{5}{3}, \frac{10}{3}\right)$

$|\overrightarrow{AC}| = \sqrt{\left(\frac{5}{3}\right)^2+\left(\frac{10}{3}\right)^2} = \sqrt{\frac{25}{9}+\frac{100}{9}} = \frac{5\sqrt{5}}{3}$

$\Rightarrow \vec{a_2}\left(-\frac{5}{3}\cdot\frac{3}{5\sqrt{5}}, \frac{10}{3}\cdot\frac{3}{5\sqrt{5}}\right)$

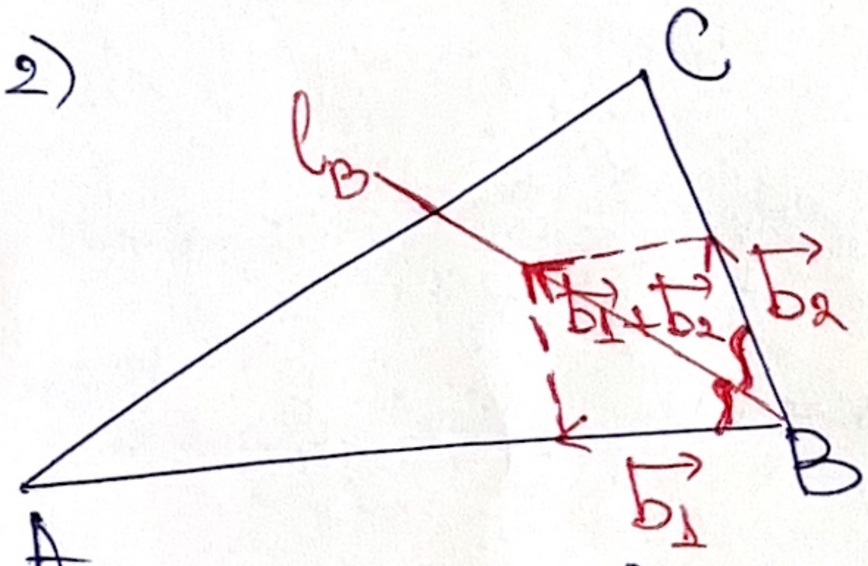
$\vec{a_2}\left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

$l_A: \begin{cases} Z_T.\ A(1,-2) \\ \parallel \vec{a_1}+\vec{a_2}\left(0, \frac{4}{\sqrt{5}}\right) \parallel \vec{e_2}(0,1) \end{cases}$

$\Rightarrow l_A: \begin{cases} x = 1+\lambda\cdot 0 \\ y = -2+\lambda\cdot 1 \end{cases}, \lambda\in\mathbb{R}$ (1H.)

(2H.) $l_A: x-1=0$

2)



$BA \parallel \overrightarrow{BA}(-1,-2) \Rightarrow \vec{b_1}\left(\frac{-1}{\sqrt{5}}, \frac{-2}{\sqrt{5}}\right)$

$|\overrightarrow{BA}| = |\overrightarrow{AB}| = \sqrt{5}$

$BC \parallel \overrightarrow{BC}\left(-\frac{8}{3}, \frac{4}{3}\right) \Rightarrow \vec{b_2}\left(-\frac{8}{3}\cdot\frac{3}{4\sqrt{5}}, \frac{4}{3}\cdot\frac{3}{4\sqrt{5}}\right)$

$|\overrightarrow{BC}| = \sqrt{\left(-\frac{8}{3}\right)^2+\left(\frac{4}{3}\right)^2} = \sqrt{\frac{16\cdot 5}{9}} = \frac{4}{3}\sqrt{5}$

$\Rightarrow \vec{b_2}\left(-\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$

$l_B: \begin{cases} Z_T.\ B(2,0) \\ \parallel \vec{b_1}+\vec{b_2}\left(-\frac{3}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right) \parallel (3,1) \end{cases}$

$l_B: \begin{cases} x = 2+\mu\cdot 3 \\ y = 0+\mu\cdot 1 \end{cases}, \mu\in\mathbb{R}$ (1H.)

(2H.) $l_B: x-3y-2=0$

2) $l_A \cap l_B = \{I\}$ - u. pe pl.

(1H.) $\begin{cases} 1+\lambda\cdot 0 = 2+\mu\cdot 3 \Rightarrow 1 = 2+3\mu \\ -2+\lambda\cdot 1 = 0+\mu\cdot 1 \Rightarrow -2+\lambda = \mu \end{cases}$

$1 = 2+3(-2+\lambda)$

$1 = 2-6+3\lambda$

$5 = 3\lambda \Rightarrow \lambda = \frac{5}{3}$

$\Rightarrow \begin{cases} x_I = 1+\frac{5}{3}\cdot 0 = 1 \\ y_I = -2+\frac{5}{3}\cdot 1 = -\frac{1}{3} \end{cases} \Rightarrow I\left(1, -\frac{1}{3}\right)$

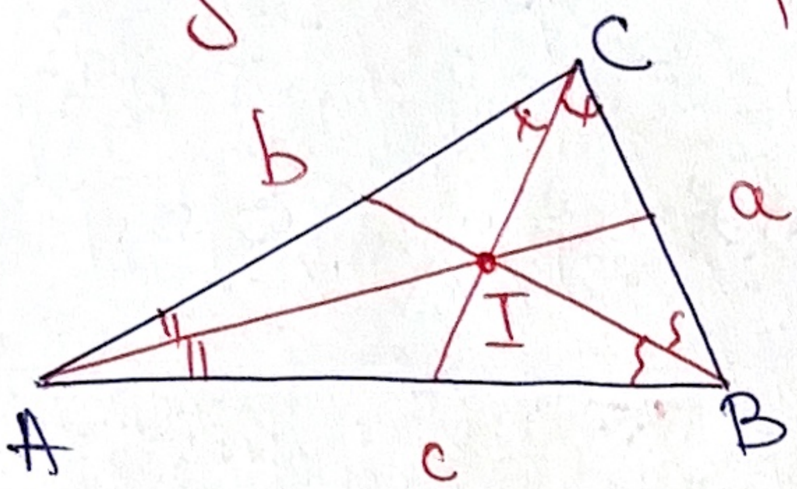
(24.) $\left|\begin{array}{l} x-3y-2=0 \\ x-1=0 \Rightarrow x=1 \end{array}\right.$ $\Rightarrow 1-3y-2=0 \Rightarrow 3y=-1 \Rightarrow y=-\frac{1}{3}$

$\Rightarrow I\left(1,-\frac{1}{3}\right)$

Алтернативно решение:

(Осн. зад.) ΔABC, $AB=c$, $BC=a$, $CA=b$

I-ч. нав. вп. окр. в ΔABC, т. O - произволна



$$\boxed{\overrightarrow{OI}=\frac{1}{a+b+c}(a.\overrightarrow{OA}+b.\overrightarrow{OB}+c.\overrightarrow{OC})}$$

В нашата задача: $c=AB=|\overrightarrow{AB}|=\sqrt{5}$

$a=BC=|\overrightarrow{BC}|=\frac{4}{3}\sqrt{5}$

$b=AC=|\overrightarrow{AC}|=\frac{5\sqrt{5}}{3}$

$a+b+c=\frac{4\sqrt{5}}{3}+\frac{5\sqrt{3}}{3}+\sqrt{5}=4\sqrt{5}$

$\overrightarrow{OI}=\frac{1}{4\sqrt{5}}(\underbrace{\frac{4}{3}\sqrt{5}}_{a''}.\overrightarrow{OA}+\underbrace{\frac{5\sqrt{5}}{3}}_{b''}\overrightarrow{OB}+\underbrace{\sqrt{5}}_{c''}.\overrightarrow{OC})$ от осн. зад.

$\underbrace{4\sqrt{5}}_{a+b+c}$

$\Rightarrow \left|\begin{array}{l} x_I=\frac{1}{4\sqrt{5}}\left(\frac{4}{3}\sqrt{5}.1+\frac{5\sqrt{5}}{3}.2+\sqrt{5}.\left(-\frac{2}{3}\right)\right) \\ y_I=\frac{1}{4\sqrt{5}}\left(\frac{4}{3}.\sqrt{5}.(-2)+\frac{5\sqrt{3}}{3}.0+\sqrt{5}.\frac{4}{3}\right) \end{array}\right.$

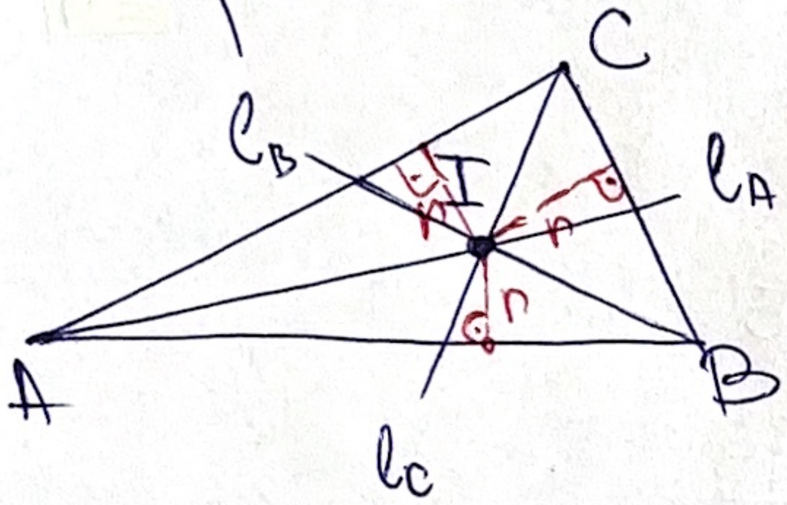
$\Rightarrow \left|\begin{array}{l} x_I=\frac{1}{\cancel{4\sqrt{5}}}.\cancel{4\sqrt{5}}=1 \\ y_I=\frac{1}{\cancel{4\sqrt{5}}}\frac{\overset{-1}{\cancel{(-4)}}.\cancel{\sqrt{5}}}{3}=-\frac{1}{3} \end{array}\right.\Rightarrow I\left(1,-\frac{1}{3}\right)$

Идея за $r=?$

$\delta(M_0,g)=\frac{Ax_0+By_0+C}{\sqrt{A^2+B^2}}$ → ориентирано разстояние от т. $M(x_0,y_0)$ до правата: $Ax+By+C=0$

$\Rightarrow \boxed{r=|\delta(I,g)|}$

В нашата задача:
(r-разстоянието от т. I до кой и да от страните на $\triangle ABC$)



$AB:\begin{cases} \in \text{т. } B(2,0) \\ \in \text{т. } A(1,-2) \end{cases}$

$AB:\begin{cases} \in \text{т. } B(2,0) \\ \parallel \overrightarrow{AB}(1,2) \end{cases} \Rightarrow$

$AB: 2x-y-4=0$

$r=\frac{|2.1-(-\frac{1}{3})-4|}{\sqrt{2^2+(-1)^2}}=\frac{|2+\frac{1}{3}-4|}{\sqrt{5}}=\frac{|\frac{1}{3}-2|}{\sqrt{5}}=\frac{5}{3\sqrt{5}}=\frac{\sqrt{5}}{3}$